Project proposal

Development of a Multimodal AI System for Detecting Deceptive Design Patterns in Web Applications

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The project proposes a web application that helps designers, developers and QA reviewers inspect suspected deceptive interface patterns. It will combine screenshots, visible text and Document Object Model features, then show evidence behind each flag for human review.

The aim is to implement and evaluate a bounded multimodal review workflow, comparing it with text-only and image-only alternatives on websites excluded from training. A gain from combining modalities is a research question, not an assumed result.

The October submission comprises analysis, design, an offline demonstration and an evaluation plan. Dataset collection, trained-model results and a working capture service are later deliverables.

Owner	Responsibilities
Zaki	Interface, capture worker, API, integration, accessibility and system testing; independent review of a dataset sample.
Akbar	Annotation protocol, dataset, baselines, multimodal inference, experiment records and error analysis.
Both	User research, literature, scope decisions, peer review and defense preparation.

Objectives and scope

Multimodal recognition already appears in prior research [R04, R05, R09]. The engineering contribution is a focused web workflow with traceable evidence and reproducible experiments, rather than a claim to be the first multimodal detector.

Initial scope: English-language public pages, one captured desktop viewport, four observable cue categories and a reviewer who makes the final judgement. Regions may carry multiple labels. An empty label set does not establish that the whole page is free of deceptive design.

Objective	Completion evidence
O1 Define the problem	3–5 real user discussions or substantive client agreement, findings and prioritised requirements.
O2 Prepare data	400–600 labelled regions including benign examples, deduplication and at least 20% independent review.
O3 Compare models	Text-only, image-only and multimodal runs using identical website-separated partitions.
O4 Integrate review	Target validation, capture, inference, evidence review and JSON export.
O5 Evaluate outcomes	Per-label precision/recall/F1, macro-F1, benign false positives, runtime and usability.

- Included: confirmshaming, misleading choice wording, visual interference and urgency cues.
- Excluded: logins, purchases, form submissions, cancellation journeys, mobile apps, legal compliance decisions and exhaustive crawling.
- Failed captures and unsupported inputs must be explicit rather than displayed as clean results.
- Dataset size and success thresholds are proposed targets for supervisor confirmation after a pilot.

Evidence of relevance and user research

Published studies establish a research problem and limits of existing detection [R01, R08]. They do not establish demand for this particular application.

No completed interview or client agreement is recorded in this pack. Collect 3–5 genuine discussions with developers, designers and QA reviewers, or obtain a substantive agreement with an actual client. Record what changes as a result.

Ask for voluntary participation and permission for anonymised notes. Use participant codes in public summaries and keep identifiable notes private.

Prompt	Evidence to record
Describe your latest interface review	Actual workflow, tools and a concrete difficulty.
What makes a flag worth investigating?	Evidence needed and examples of false alarms.
Try the supplied demonstration	Observed confusion, missing information and useful actions.
What result would you act on?	Acceptance conditions and responsibility for decisions.
What should this tool avoid?	Privacy and scanning restrictions.

- Record date, role, consent, observations, authorised short quotes, findings and linked requirement IDs.
- Unvalidated hypotheses: users need region evidence (FR03), editable judgements (FR04) and export (FR05).
- Use planning/user_research_template.md. Preserve negative feedback and disagreements.

Literature review and research gap

Twelve research sources inform the design: eleven identified conference or journal publications and one clearly labelled preprint. Ten date from 2022 onward. The 2019 study and 2021 conceptual analysis are retained as foundations. Source pages were checked on 23 October 2026.

Definitions precede models. The conceptual analysis [R02] and ontology [R03] support a small operational label guide that separates visible cues from motive. Overlapping concepts justify multi-label annotation and an uncertainty state.

Text baselines provide a reproducible reference [R06]. Screenshot and UI-property approaches [R04, R05] motivate image localisation and feature extraction. This project will compare models on identical grouped splits rather than rank metrics reported on incompatible datasets.

Newer multimodal work [R09] prevents novelty claims based only on combining images and text. The proposed contribution links screenshot evidence, DOM features, model versions and reviewer decisions in a narrow web workflow.

Coverage remains uneven [R07, R08]. Account disabling and cancellation [R10, R12] depend on sequences that one screenshot cannot capture. Those behaviours remain outside scope. Highlighting tools [R11] motivate separate evaluation of reviewer usefulness.

This is a targeted review, not an exhaustive systematic review. Searches covered definitions, datasets, screenshot recognition, multimodal detection and interaction-dependent limitations. The tables show each source’s approach, limitations and concrete use.

Publication analysis 1

Source	Approach	Limitation and use
R01 Mathur et al. 2019 PACM HCI, CSCW	Large web crawl with expert categorisation.	Shopping and text emphasis restrict transfer to visual patterns. Foundation for urgency and confirmshaming labels, not current prevalence.
R02 Mathur, Mayer and Kshirsagar 2021 CHI 2021	Connects interface attributes with different concepts of harm.	A conceptual framework does not provide classifier ground truth. Separate visible cues from conclusions about motive or wrongdoing.
R03 Gray, Bielova, Santos and Mildner 2024 CHI 2024	Harmonises ten taxonomies into 64 types at three levels.	Broad concepts need operational annotation rules. Map four project labels to established concepts and document overlaps.
R04 Mansur, Salma, Awofisayo and Moran 2023 ICSE 2023	Combines visual and text cues for screenshot detection and localisation.	Performance varies across types and benchmark coverage is limited. Return region evidence and report class-level results.

Publication analysis 2

Source	Approach	Limitation and use
R05 Chen et al. 2023 UIST 2023; manuscript revised 2024	UIGuard extracts UI properties and applies knowledge-driven checks.	Mobile results do not establish website performance. Use property features and benign examples, then validate on web domains.
R06 Yada et al. 2022 IEEE BigData 2022	Adds negative text examples and compares language models.	Text cannot represent purely visual choice imbalance. Include text-only baselines and explicit benign examples.
R07 Soe, Santos and Slavkovik 2022 Research preprint; publication status unconfirmed	Studies engineered features and a classifier for cookie banners.	Narrow context and annotation ambiguity limit transfer. Keep uncertainty and adjudication in the label protocol.
R08 Nie, Zhao, Li, Luo and Liu 2024 PACM Software Engineering, FSE	Reviews taxonomies, tools and datasets.	Detection coverage and available data remain uneven. Distinguish unsupported categories from negative findings.

Publication analysis 3

Source	Approach	Limitation and use
R09 Shi et al. 2025 The Web Conference 2025	DPGuard combines multimodal reasoning and an expanded taxonomy.	Commercial model dependence complicates cost and repeatability. Acknowledge prior multimodal work and compare local reproducible baselines.
R10 Kelly and Rubin 2024 Social Media + Society	Analyses account-disabling sequences on social networking sites.	The behaviours depend on interaction history. Exclude cancellation obstruction from a single-page prototype.
R11 Donnelly et al. 2022 ETHICOMP 2022	Proposes a framework and tool to detect and highlight web patterns.	A framework is not proof of performance for the proposed scope. Evaluate reviewer usefulness separately from model accuracy.
R12 Runge, Wentzel, Huh and Chaney 2022 online; 2023 issue Marketing Letters 34, 155–160	Observes cancellation practices and discusses incentives.	Incentives cannot be inferred from a screenshot. Avoid claiming motive and record limits of single-page evidence.

Existing system comparison

Four published research systems are compared using eight common criteria. Entries reflect cited descriptions, not independent reproduction. “Not established” means the reviewed source does not establish the feature. The proposal has not been benchmarked against these tools.

Criterion	Mathur [R01]	AidUI [R04]	UIGuard [R05]	DPGuard [R09]
Setting	Shopping web	Web/mobile screenshots	Mobile apps	Web/mobile UIs
Input	Crawled web text	Screenshot cues	UI properties and text	Multimodal UI input
Approach	Collection and expert analysis	Vision plus text cues	Knowledge-driven checks	Multimodal large-model reasoning
Output	Mined instances	Types and localised regions	Detected types	Detected patterns
Benign data	Not a matched classifier here	Benchmark-specific	Explicit benign UI data	Classifier stage described
Browser DOM fusion	Not proposed fusion	Not established	Mobile rather than web DOM	Not established
Multi-step coverage	Not comprehensive	Screenshot limitation	Screen context emphasis	Depends on supplied context
Transfer limit	Shopping/text scope	Category and data coverage	Mobile-to-web transfer	Service and prompt dependence

- Proposed distinction: a local four-label web review workflow with aligned screenshot/text/DOM evidence and reviewer corrections.
- Published accuracy figures use different tasks and datasets, so they are not presented as a league table.
- Implementation and evaluation are required before claiming an advantage.

Technology decisions

Three decisions compare three alternatives each. Selections are proposed and require a feasibility spike. Criteria are requirement fit, integration effort and operational complexity, not unmeasured speed claims.

Decision	Alternatives and tradeoffs	Proposed choice
API platform	FastAPI: Python schemas/validation [T01]. Django: integrated ORM/admin [T02]. Express: JavaScript middleware [T03], requiring a Python inference boundary.	FastAPI for direct Python integration. FR01–FR06.
Storage	SQLite: embedded local store [T04]. PostgreSQL: shared relational service [T05]. MongoDB: flexible documents [T06] with explicit application validation.	SQLite for one-machine pilot; private image files outside DB. Revisit PostgreSQL for concurrent use.
Capture	Playwright: Python and isolated contexts [T07]. Selenium: WebDriver ecosystem [T08]. Puppeteer: JavaScript browser control [T09], adding a runtime boundary.	Playwright in isolated worker with fixed viewport, bounded loads and request restrictions.

- React and TypeScript are planned for the interface. The deadline demonstration is dependency-free HTML.
- Begin with fixed pretrained embeddings and lightweight classifier heads. Defer fine-tuning until a dataset pilot supports it.
- Queue capture and inference jobs, return a scan ID and poll status instead of blocking a long HTTP request.

Functional requirements 1

These are acceptance criteria for the future implemented system. The mockup demonstrates interaction and does not satisfy capture or model requirements.

ID / priority	Behaviour and acceptance	Owner / check
FR01 Must	Submit a target. Reject malformed URLs, credentials, non-HTTPS targets and hosts outside the configured allowlist. Valid requests return scan ID and queued status.	Zaki V01
FR02 Must	Capture evidence. Save 1366×768 screenshot, visible text, region coordinates, permitted DOM features, timestamp and viewport. Capture failures produce an error state.	Zaki V02
FR03 Must	Show findings. Display supported labels, evidence, model score and model version for each region. Scores are not labelled as calibrated probabilities.	Both V03
FR04 Must	Record review. Confirm, dismiss or mark uncertain with a note. Preserve original model output separately from the reviewer decision.	Zaki V04
FR05 Must	Export report. JSON includes target, status, timestamp, version, evidence references and review decisions. Failed scans remain explicit.	Zaki V05

Functional requirements 2

These are acceptance criteria for the future implemented system. The mockup demonstrates interaction and does not satisfy capture or model requirements.

ID / priority	Behaviour and acceptance	Owner / check
FR06 Must	Track progress. Show queued, capturing, analysing, completed or failed. Timeout errors are visible; retries create new IDs.	Zaki V06
FR07 Must	Run experiments. Record split manifests, configuration, seed, dependency/model versions and per-label metrics for three modality conditions.	Akbar V07
FR08 Must	Track annotation. Every accepted example has provenance, coordinates, labels, annotator, review state and site group. Independently review at least 20%.	Akbar V08
FR09 Should	Filter results. Filter by label and review state, preserving data and showing a useful empty state.	Zaki V09
FR10 Could	Compare runs. Compare two versioned model outputs on the same saved capture. Defer if Must requirements need time.	Both V10

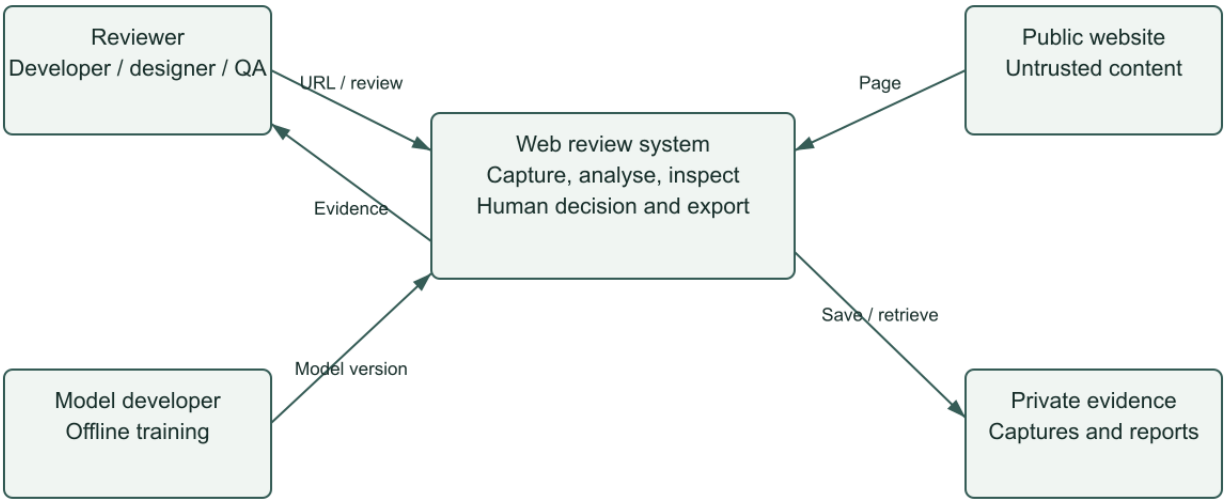
Non functional requirements

ID / priority	Condition	Check
NFR01 Must	Capture isolation. Check top-level navigation, redirects and subrequests against approved hosts and public destination IPs. Block private, loopback, link-local and metadata addresses. Disable downloads and form actions. Use an isolated worker and OS/container egress controls.	V11
NFR02 Must	Bounded runtime. Proposed target: scan within 60 seconds on declared hardware. Enforce 30-second navigation and 90-second total job limits. Report p50/p95 and failures for 20 controlled scans.	V12
NFR03 Must	Data minimisation. Public pages without login. Exclude input values, cookies and credentials. Keep raw captures out of Git and delete after a proposed 30-day retention period.	V13
NFR04 Must	Traceability. Every export has capture ID, model version and timestamp; original predictions remain separate from review decisions.	V14
NFR05 Must	Accessibility. Submit, review and export by keyboard. Label inputs, show focus, announce status and use text as well as colour.	V15
NFR06 Must	Reproducibility. Pin dependencies, seeds and partitions. Record rerun variation rather than promise bit-identical GPU output.	V07
NFR07 Should	Maintainability. Separate API, capture, inference and UI; link every Must requirement to a check and owned task.	V16

System context

The reviewer submits an approved public page and receives evidence for a human decision. Target websites are external, untrusted data sources. The model developer supplies versioned artefacts from an offline workflow.

The proposed first deployment is a local pilot operated by the team. A public multi-user service and production hosting are outside the first milestone. Capture isolation remains necessary when running locally.

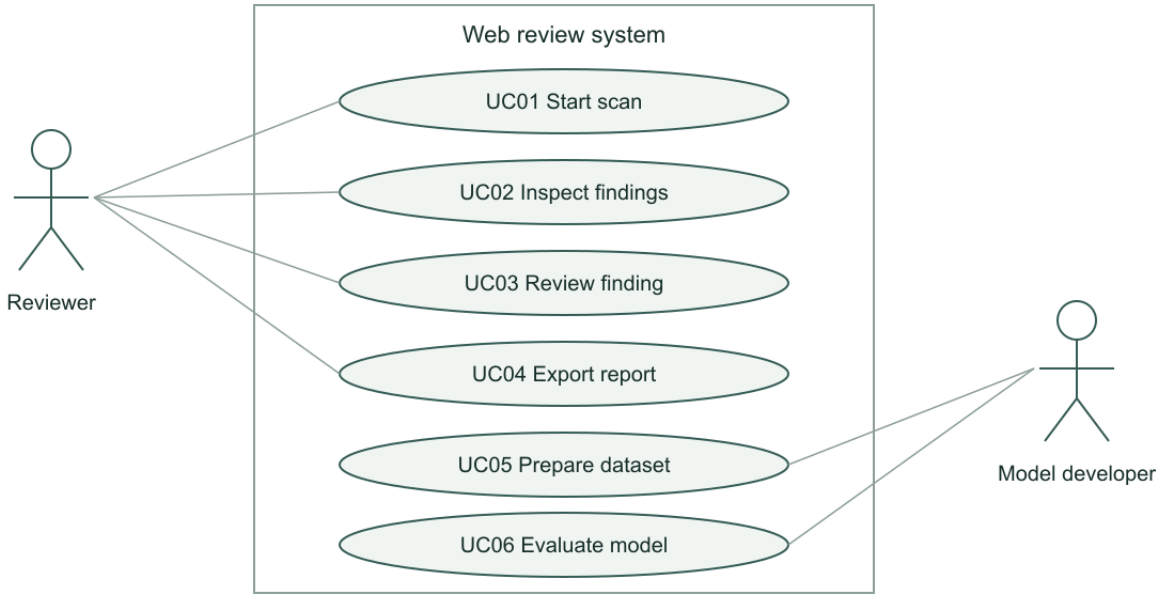


- FR01–FR06 support review. FR07–FR08 support data and model development.
- The demonstration uses synthetic fixtures, not live website captures.

Use cases

The reviewer starts a scan, checks progress, inspects evidence, records a judgement and exports a report. Invalid or unsupported targets lead to an explicit error. A retry creates a new scan.

The model developer prepares labelled data, evaluates models and publishes a versioned inference artefact. Supervision and peer review are project activities rather than application use cases.



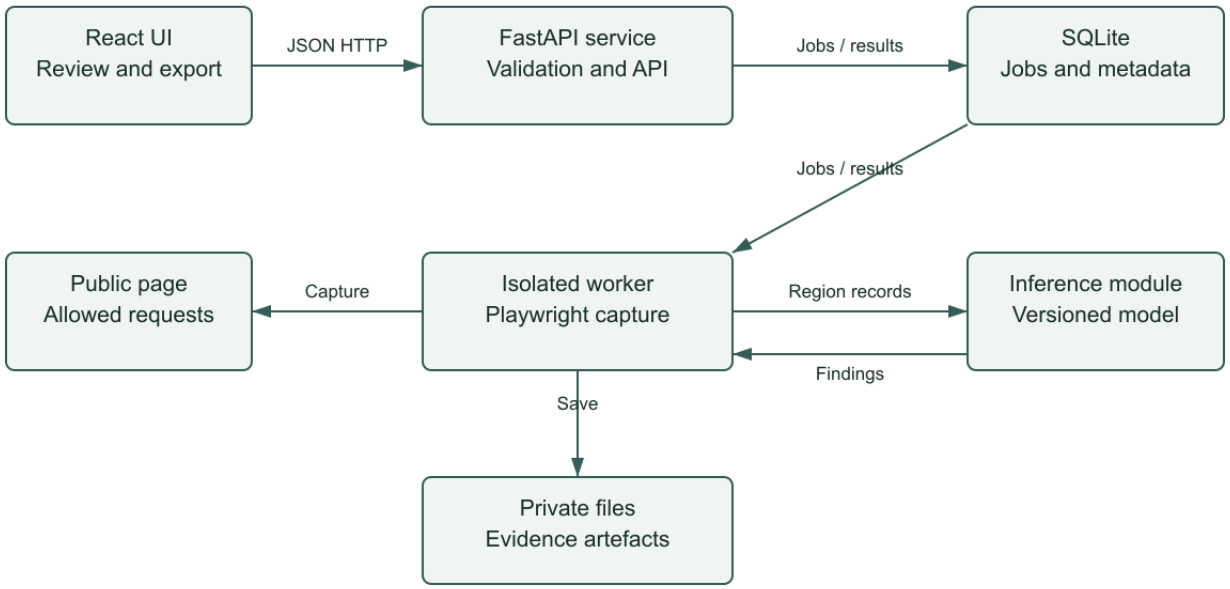
- UC01 Start scan includes validation. UC02 Inspect findings requires completed analysis.
- UC03 Review preserves model output. UC04 Export includes the reviewer state.
- UC05 Prepare dataset and UC06 Evaluate model implement FR07–FR08.

Component architecture

The React interface calls FastAPI using JSON HTTP. The API validates targets and creates jobs in a persistent SQLite table. A local worker claims jobs, captures an isolated page and calls the inference module with normalised region records.

The pilot uses one worker and one API instance. Claim jobs atomically, use leases and bounded retries, and recover expired leases or mark jobs failed after restart. A dedicated queue and shared database can follow if deployment needs expand.

Private files store evidence; SQLite stores metadata and references. Inference returns supported labels, scores and evidence links. Model training runs offline.



Data and API contract

Capture records contain scan_id, approved target, timestamp, viewport, evidence references and regions. Each region includes region_id, bounding box in CSS pixels, visible text, selected DOM properties and zero or more supported labels. Review records preserve the original model result.

contracts/openapi.json specifies the planned boundary. Examples are synthetic. Failure and unsupported cases must not become successful clean results.

Endpoint	Purpose	Result
POST /scans	Validate and queue	202 ID; 422 invalid target
GET /scans/{scan_id}	Progress and results	200 state; 404 unknown scan
PATCH /scans/{scan_id}/findings/{finding_id}/review	Save reviewer decision	200 saved; 404 unknown; 422 invalid
GET /scans/{scan_id}/report	Export completed scan	200 JSON; 409 not completed

- Job states: queued, capturing, analysing, completed, failed. A retry is a separate job.
- Multi-label outputs allow overlapping cues. Scores are not proof of deception or verified probabilities.
- Reviewer state is confirmed, dismissed or uncertain. It must not overwrite predictions.

Dataset and annotation plan

Target 400–600 regions from at least 30 registrable domains if feasible, including a substantial benign set. Begin with a 50-region pilot. These are plans, not collected data.

Record provenance, reuse conditions and capture settings. Public accessibility does not establish redistribution rights. Keep raw third-party captures outside the public repository until reuse and redaction are checked.

Label	Operational rule	Boundary
Confirmshaming	Refusal option uses degrading or guilt-inducing language.	Neutral “No thanks” alone is negative.
Misleading choice wording	Wording or polarity obscures the consequence of selection.	Require surrounding context that explains the ambiguity.
Visual interference	One available choice is materially harder to notice.	Record relative size, placement and contrast; colour alone is insufficient.
Urgency cue	Countdown or time pressure encourages immediate action.	Does not prove the time limit is false.
No supported cue	None of the four labels is supported.	Does not certify the entire page.

- Akbar annotates; Zaki independently reviews at least 20% before seeing initial labels. Record raw agreement and per-label Cohen’s kappa where defined, then adjudicate.
- Mark ambiguous examples uncertain and exclude from the primary scored set or report separately.
- Remove exact/near duplicates before splitting. Group by registrable domain and shared template family where known.
- Aim for 60/20/20 group splits with class coverage. Freeze manifests before evaluation.

Model experiments and evaluation

RQ1 Does multimodal fusion improve macro-F1 over the best unimodal model on unseen websites? RQ2 Which labels and conditions cause false positives? RQ3 Can reviewers interpret evidence and complete the workflow?

Use identical accepted samples and website-grouped partitions. Use TF-IDF plus a linear classifier as a simple text baseline. For the matched comparison, use the same fixed pretrained text and image encoders in all conditions: text-only, image-only, and concatenated text/image embeddings with normalised DOM features. Train comparable lightweight heads per label. Record encoder identity, licence and versions when selected.

Tune only on validation data. Fit vocabularies, scalers and dimensionality reduction only on training data. Select configuration before held-out testing. If necessary, use grouped cross-validation within training while preserving a final test set.

Metric	Definition
Per-label precision/recall	TP/(TP+FP) and TP/(TP+FN); report support and zero-denominator policy.
Macro-F1	Mean F1 across the four labels, accompanied by all per-label values.
Benign false-positive rate	Benign regions with any predicted label / all benign regions. Report counts.
Fusion improvement	Multimodal macro-F1 minus best unimodal macro-F1 on the same test split. Bootstrap by website where feasible.
Runtime and coverage	p50/p95, capture failures, unsupported inputs and declared hardware.

- Provisional targets: macro-F1 at least 0.70 and benign false-positive rate at most 0.15. These are planning targets, not measured results or university rules.
- A negative fusion result is valid if methods and limitations are documented. Do not repeatedly tune to test errors.

Verification and usability plan

Checks V01–V16 in docs/verification_matrix.md define future implementation acceptance. They are not reported as passed. Prototype checks are recorded separately.

After consent, run a small formative usability exercise with 3–5 representative users. This is distinct from the initial needs discussions. Ask users to inspect one flag, dismiss a false alarm, export a result and explain the limitations.

Record task completion, time, mistakes and interpretation. Proposed goal: at least 80% task completion and correct recognition that flags require human review. Report actual counts and the small sample.

Check	Requirements	Procedure and expected evidence
V01	FR01	Allowed, malformed, credential-bearing and blocked URLs: Only valid approved targets create jobs.
V02	FR02	Known capture fixture: Screenshot, coordinates and extracted text agree.
V04	FR04	Confirm/dismiss/uncertain then reload: Review persists; original prediction is unchanged.
V05	FR05	Export and parse JSON: Required fields and review state match the screen.
V07	FR07/NFR06	Versioned experiment rerun: Same split manifest, versions, seeds and recorded metrics.
V08	FR08	Manifest and annotation audit: Complete records, no split leakage and at least 20% independent review.
V11	NFR01	Private IP, redirect, subresource and DNS changes: Network policy blocks disallowed destinations.

Prototype demonstration

Open prototype/index.html in a browser. It requires no installation or network connection. It demonstrates target entry, progress, highlighted evidence, review decisions, filtering and JSON export using synthetic examples.

The interface labels all example findings and scores as illustrative. It does not fetch URLs, run an AI model or establish detection accuracy.

- Choose the fictional shop, run the demonstration and inspect confirmshaming and urgency cues.
- Dismiss a finding with a note, filter review state and export the JSON result.
- Choose the neutral fixture and inspect the qualified no-supported-cues message.
- Choose the capture-failure fixture and demonstrate recovery.
- Use the demonstration to gather real feedback on wording and evidence requirements.
- Backend capture, model training and persistent server storage remain implementation work.

Management and milestones

The proposed twelve-week development window is 26 October 2026 to 17 January 2027. The university final defense date is not recorded; the supervisor must align this schedule before it becomes binding. Weekly meetings and fortnightly integration reviews follow the proposed supervision plan.

The repository inspected on 23 October contained one initial commit and a short README. This establishes a starting point, not progressive contributions by both students. Each student must understand, review and commit genuine subsequent work using their own account.

Period	Zaki	Akbar	Checkpoint
24 October	Submission and UX demo	Research and model-plan review	Supervisor comments and evidence gaps
Weeks 1–4	Capture, API and dashboard	Pilot, dataset and splits	Feasibility and requirements
Weeks 5–8	Review flow and integration	Baselines, fusion and inference	Integration and frozen protocol
Weeks 9–12	System tests and usability	Held-out evaluation and errors	Final evidence and defense

- planning/task_backlog.json and .md contain owners, priorities, dates and acceptance criteria.
- P01–P14 are local planning labels, not actual YouTrack IDs. Create real tasks and replace/link the IDs.
- Use task branches and pull requests, with another team member reviewing before merge.

Risks and mitigations

Risk	Likelihood / impact	Response	Owner
Missing user or supervisor evidence	High / high	Keep pending and obtain real dated records.	Both
Small or ambiguous dataset	High / high	50-region pilot, independent review and scope reduction if necessary.	Akbar
Website leakage	Medium / high	Deduplicate and group by domain/template before splitting.	Akbar
Unsafe page requests	Medium / high	Allowlist, request checks, isolated worker and network egress policy.	Zaki
Capture mismatch	Medium / medium	Fixed viewport, browser metadata and coordinate tests.	Zaki
Compute or time constraints	Medium / high	Fixed encoders and small heads; defer optional features.	Akbar
Overstated model conclusions	Medium / high	Suspected-cue wording, evidence and human review.	Both
Sensitive material in public Git	Medium / high	Private raw data; publish only permitted anonymised summaries.	Both
Unknown final deadline	Medium / high	Confirm university schedule and revise dates.	Both

Submission checklist and supervision

Prepared documents support analysis and design. They do not prove that interviews, supervision, peer review or experiments have occurred.

Area	Prepared	Remaining action
Proposal	Aim, scope, roles and objectives	Team and supervisor confirmation
Relevance	Research synthesis and interview form	Real discussions or client agreement
Literature	12-source analysis and bibliography	Team verifies understanding and relevant full texts
Comparisons	4 systems × 8 criteria; 3 technology decisions	Feasibility checks
Requirements and architecture	Acceptance criteria and 3 diagrams	Peer and supervisor review
UX and evaluation	Offline demo and verification plan	Actual usability and implementation evidence later
Management	Owned backlog, risks and schedule	Real YouTrack project, assignments and access
Supervision	Feedback form	Actual comments and revisions
Repository	Proposed package and workflow	Genuine contributions, review and accepted submission tag

- Confirm the submission time and channel with the university.
- Before tagging, open every deliverable, resolve or disclose gaps and merge accepted work.
- Create pre-defense-1 once on the accepted commit. Never move a submitted tag.
- Feedback log fields: date, deliverable, actual comment, decision, owner, revision link and unresolved issue. No approval or signature is supplied.

Research references 1

[R01] Mathur et al. (2019). Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites. PACM HCI, CSCW. https://doi.org/10.1145/3359183

[R02] Mathur, Mayer and Kshirsagar (2021). What Makes a Dark Pattern... Dark? Design Attributes, Normative Considerations, and Measurement Methods. CHI 2021. https://arxiv.org/abs/2101.04843

[R03] Gray, Bielova, Santos and Mildner (2024). An Ontology of Dark Patterns: Foundations, Definitions, and a Structure for Transdisciplinary Action. CHI 2024. https://doi.org/10.1145/3613904.3642436

[R04] Mansur, Salma, Awofisayo and Moran (2023). AidUI: Toward Automated Recognition of Dark Patterns in User Interfaces. ICSE 2023. https://arxiv.org/abs/2303.06782

[R05] Chen et al. (2023). Unveiling the Tricks: Automated Detection of Dark Patterns in Mobile Applications. UIST 2023; manuscript revised 2024. https://arxiv.org/abs/2308.05898

[R06] Yada et al. (2022). Dark patterns in e-commerce: a dataset and its baseline evaluations. IEEE BigData 2022. https://arxiv.org/abs/2211.06543

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[R07] Soe, Santos and Slavkovik (2022). Automated detection of dark patterns in cookie banners: how to do it poorly and why it is hard to do it any other way. Research preprint; publication status unconfirmed. https://arxiv.org/abs/2204.11836

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Technical references

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[T02] Django overview. https://docs.djangoproject.com/en/5.2/intro/overview/

[T03] Express. https://expressjs.com/

[T04] Appropriate uses for SQLite. https://www.sqlite.org/whentouse.html

[T05] About PostgreSQL. https://www.postgresql.org/about/

[T06] MongoDB data modelling. https://www.mongodb.com/docs/manual/data-modeling/

[T07] Playwright for Python. https://playwright.dev/python/docs/intro

[T08] Selenium WebDriver. https://www.selenium.dev/documentation/webdriver/

[T09] Puppeteer. https://pptr.dev/

Official documentation checked 23 October 2026. Exact implementation versions will be pinned after the feasibility spike.